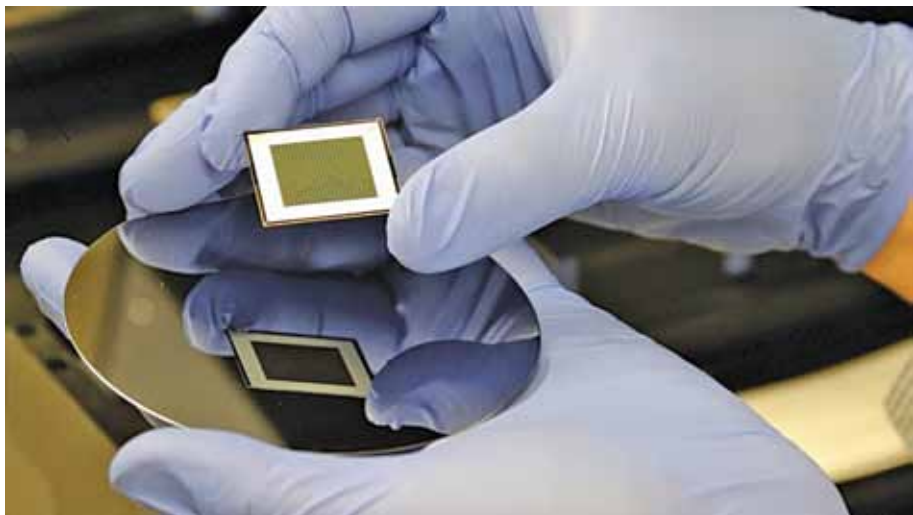




Bifacial solar cells that produce 30% more power

Scientists at The Australian National University (ANU) have produced an efficient type of solar cells using laser processing. The solar cells are dual sided, meaning both the front and back of the cell generate power. When deployed on a conventional solar farm, a bifacial cell absorbs direct incoming light, while also taking advantage of ground reflection, which can contribute up to additional 30% power generation. The team used specific laser doping technology to fabricate the cells, which allowed the research team to achieve a front conversion efficiency of 24.3% and a rear conversion efficiency of 23.4%, representing a bifacial factor of 96.3%. With these statistics, the bifacial solar cells can easily beat the performance of conventional single-sided silicon solar cells.



Bifacial solar cells (Credit: www.anu.edu.au)



AI system-on-chip that runs on solar power



System-on-chip works through an entirely new signal processing architecture that minimises the amount of power needed (Credit: www.csem.ch)

Artificial Intelligence (AI) is used in an array of extremely useful applications, but the AI based technology generally requires a lot of power and must be permanently connected to the cloud, raising issues related to data protection, IT security, and energy use. Now, engineers at the Swiss Center for Electronics and Microtechnology (CSEM) have developed a new system-on-chip to get around those issues. It runs on a tiny battery or a small solar cell and executes AI operations at the edge, that is, locally on the chip rather than in the cloud. What's more, their system is fully modular and can be tailored to any application where real-time signal and image processing is required, especially when sensitive data are involved.